

PAC

Question 3

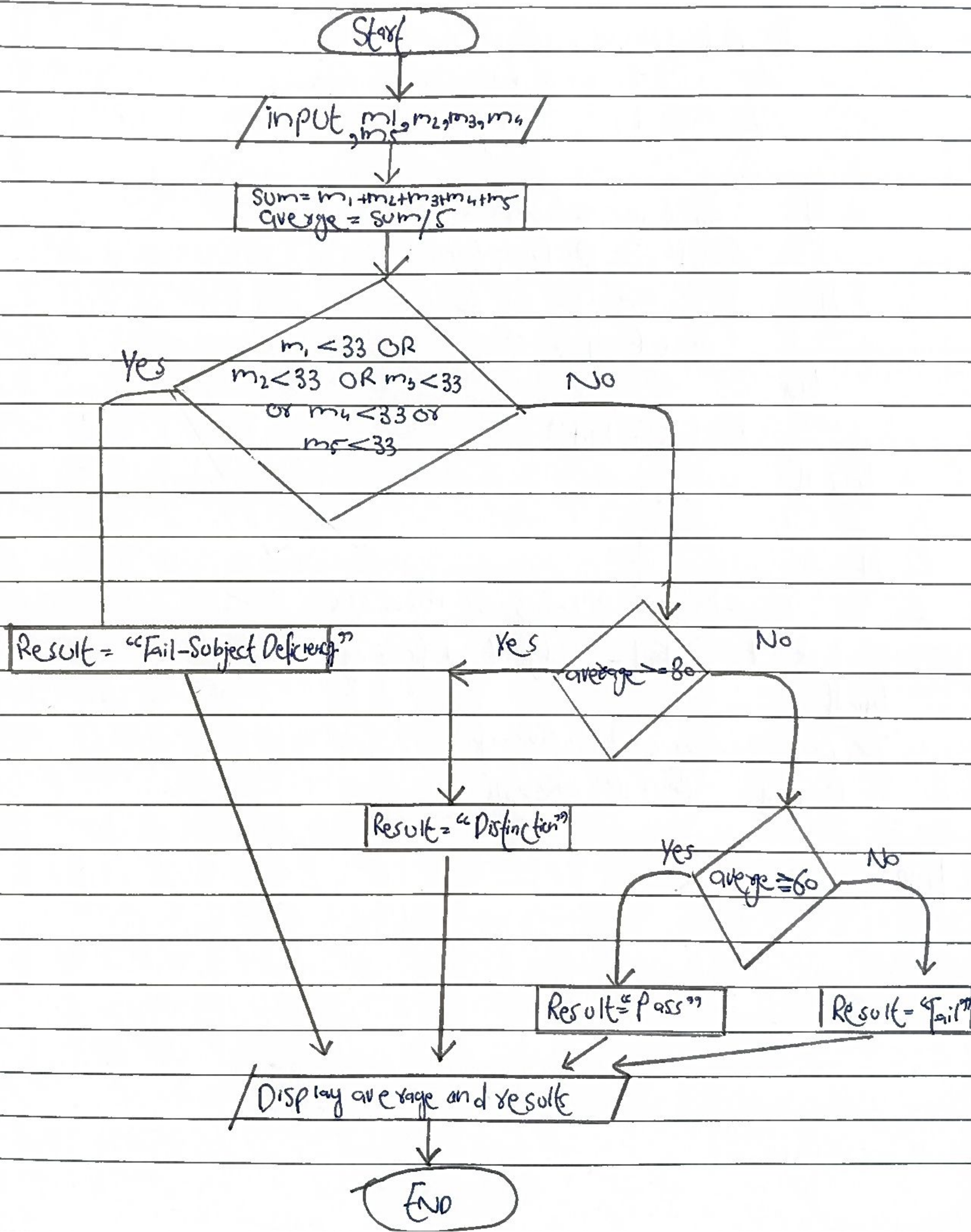
Input	Required Output	Processing
Five subject marks m_1, m_2, m_3, m_4, m_5 (out of 100 each)	Average marks final result classification	$Sum = m_1 + m_2 + m_3 + m_4 + m_5$ $\rightarrow \text{average} = Sum / 5$ $\rightarrow$ Classify by average ≥ 80 Distinction $60-79$ pass, < 60 fail $\rightarrow$ override to fail — Subject deficiency if any single mark < 33

TPO

Input	Process	Output
m_1	$\rightarrow Sum = m_1 + m_2 + m_3 + m_4 + m_5$	$\rightarrow$ AVERAGE
m_2	$\rightarrow \text{Average} = Sum / 5$	$\rightarrow$ Final result
m_3	$\rightarrow$ Nested if: $\text{average} \geq 80$	
m_4	Distinction: $\geq 60 \rightarrow$ pass	
m_5	else $\rightarrow$ fail	
	$\rightarrow$ If any mark < 33	
	override result	

to "fail - Subject deficiency"

Flowchart



Pseudo Code:-

Begin

Input m_1, m_2, m_3, m_4, m_5

Sum = $m_1 + m_2 + m_3 + m_4 + m_5$

average = $\text{Sum} / 5$

If average ≥ 80
 result = "Distinction"

Else if average ≥ 60
 result = "pass"

Else
 result = "Fail"

End If

If

$m_1 < 33$ or $m_2 < 33$ or $m_3 < 33$ or $m_4 < 33$ or $m_5 < 33$

result = "Fail - Subject deficiency"

End If

Display "average", average

Display "Result", Result

End